

Clinical Risk Assessment Report: Patient 78

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.44x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.6%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.61 [Bottom 24%] (-2.6%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.12 [Avg (42th %ile)] (-1.1%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.34 [Top 8%] (-1.0%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Patient Age (Age) **Value: 65.0 [Avg (54th %ile)] (-0.9%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.02 [Avg (44th %ile)] (-0.8%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.43 [Bottom <1%] (-0.7%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.3 [Avg (53th %ile)] (-0.5%)**

Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.42 [Avg (64th %ile)] (-0.4%)**

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Machine Learning Feature Attribution (SHAP Decision Path)

